MSG No. 207.40 $P432$ [Type I, cubic]

Table 1: Wyckoff site: `1a`, site symmetry: `432`

No.	position	mapping
1	[0, 0, 0]	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]`

Table 2: Wyckoff site: `1b`, site symmetry: `432`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]`

Table 3: Wyckoff site: `3c`, site symmetry: `42.2`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,3,8,9,10,13,14]`
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[4,5,15,16,18,19,21,23]`
3	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[6,7,11,12,17,20,22,24]`

Table 4: Wyckoff site: `3d`, site symmetry: `42.2`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	`[1,2,3,8,9,10,13,14]`
2	$\left[0, 0, \frac{1}{2}\right]$	`[4,5,15,16,18,19,21,23]`
3	$\left[0, \frac{1}{2}, 0\right]$	`[6,7,11,12,17,20,22,24]`

Table 5: Wyckoff site: `6e`, site symmetry: `4..`

No.	position	mapping
1	$[x, 0, 0]$	`[1,2,3,8]`
2	$[0, 0, -x]$	`[4,16,19,23]`
3	$[0, 0, x]$	`[5,15,18,21]`
4	$[0, x, 0]$	`[6,11,17,24]`
5	$[0, -x, 0]$	`[7,12,20,22]`
6	$[-x, 0, 0]$	`[9,10,13,14]`

Table 6: Wyckoff site: `6f`, site symmetry: `4..`

No.	position	mapping
1	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,3,8]`
2	$\left[\frac{1}{2}, \frac{1}{2}, -x\right]$	`[4,16,19,23]`
3	$\left[\frac{1}{2}, \frac{1}{2}, x\right]$	`[5,15,18,21]`
4	$\left[\frac{1}{2}, x, \frac{1}{2}\right]$	`[6,11,17,24]`
5	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	`[7,12,20,22]`

continued ...

Table 6

No.	position	mapping
6	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	[9,10,13,14]

Table 7: Wyckoff site: 8g, site symmetry: .3.

No.	position	mapping
1	$[x, x, x]$	[1,17,18]
2	$[x, -x, x]$	[2,7,15]
3	$[x, x, -x]$	[3,4,11]
4	$[-x, x, x]$	[5,6,13]
5	$[x, -x, -x]$	[8,22,23]
6	$[-x, x, -x]$	[9,19,24]
7	$[-x, -x, x]$	[10,20,21]
8	$[-x, -x, -x]$	[12,14,16]

Table 8: Wyckoff site: 12h, site symmetry: 2..

No.	position	mapping
1	$\left[x, \frac{1}{2}, 0\right]$	[1,8]
2	$\left[x, 0, \frac{1}{2}\right]$	[2,3]
3	$\left[0, \frac{1}{2}, -x\right]$	[4,16]
4	$\left[0, \frac{1}{2}, x\right]$	[5,15]
5	$\left[\frac{1}{2}, x, 0\right]$	[6,11]
6	$\left[\frac{1}{2}, -x, 0\right]$	[7,12]
7	$\left[-x, \frac{1}{2}, 0\right]$	[9,10]
8	$\left[-x, 0, \frac{1}{2}\right]$	[13,14]
9	$\left[0, x, \frac{1}{2}\right]$	[17,24]
10	$\left[\frac{1}{2}, 0, x\right]$	[18,21]
11	$\left[\frac{1}{2}, 0, -x\right]$	[19,23]
12	$\left[0, -x, \frac{1}{2}\right]$	[20,22]

Table 9: Wyckoff site: 12i, site symmetry: ..2

No.	position	mapping
1	$[0, y, y]$	[1,13]
2	$[0, -y, y]$	[2,10]
3	$[0, y, -y]$	[3,9]
4	$[y, y, 0]$	[4,18]
5	$[-y, y, 0]$	[5,19]
6	$[-y, 0, y]$	[6,20]
7	$[y, 0, y]$	[7,17]
8	$[0, -y, -y]$	[8,14]
9	$[y, 0, -y]$	[11,22]
10	$[-y, 0, -y]$	[12,24]

continued ...

Table 9

No.	position	mapping
11	$[y, -y, 0]$	`[15,23]`
12	$[-y, -y, 0]$	`[16,21]`

Table 10: Wyckoff site: `12j`, site symmetry: `..2`

No.	position	mapping
1	$\left[\frac{1}{2}, y, y\right]$	`[1,13]`
2	$\left[\frac{1}{2}, -y, y\right]$	`[2,10]`
3	$\left[\frac{1}{2}, y, -y\right]$	`[3,9]`
4	$\left[y, y, \frac{1}{2}\right]$	`[4,18]`
5	$\left[-y, y, \frac{1}{2}\right]$	`[5,19]`
6	$\left[-y, \frac{1}{2}, y\right]$	`[6,20]`
7	$\left[y, \frac{1}{2}, y\right]$	`[7,17]`
8	$\left[\frac{1}{2}, -y, -y\right]$	`[8,14]`
9	$\left[y, \frac{1}{2}, -y\right]$	`[11,22]`
10	$\left[-y, \frac{1}{2}, -y\right]$	`[12,24]`
11	$\left[y, -y, \frac{1}{2}\right]$	`[15,23]`
12	$\left[-y, -y, \frac{1}{2}\right]$	`[16,21]`

Table 11: Wyckoff site: `24k`, site symmetry: `1`

No.	position	mapping
1	$[x, y, z]$	`[1]`
2	$[x, -z, y]$	`[2]`
3	$[x, z, -y]$	`[3]`
4	$[z, y, -x]$	`[4]`
5	$[-z, y, x]$	`[5]`
6	$[-y, x, z]$	`[6]`
7	$[y, -x, z]$	`[7]`
8	$[x, -y, -z]$	`[8]`
9	$[-x, y, -z]$	`[9]`
10	$[-x, -y, z]$	`[10]`
11	$[y, x, -z]$	`[11]`
12	$[-y, -x, -z]$	`[12]`
13	$[-x, z, y]$	`[13]`
14	$[-x, -z, -y]$	`[14]`
15	$[z, -y, x]$	`[15]`
16	$[-z, -y, -x]$	`[16]`
17	$[z, x, y]$	`[17]`
18	$[y, z, x]$	`[18]`
19	$[-y, z, -x]$	`[19]`
20	$[-z, -x, y]$	`[20]`
21	$[-y, -z, x]$	`[21]`
22	$[z, -x, -y]$	`[22]`

continued ...

Table 11

No.	position	mapping
23	$[y, -z, -x]$	`[23]`
24	$[-z, x, -y]$	`[24]`